Module 4: Sampling and Estimation

307312 Applied Statistics | University of Petra

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Module 4: Sampling and Estimation

Four lessons | From sample selection to uncertainty

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Learning sequence

Complete Module 3 first. We distinguish individual outcomes from sample statistics, develop standard errors, then use them to construct intervals and plan precision.

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1. Sampling and Variability	1. Practice
2. Sampling Distributions and Standard Errors	2. Practice
3. Confidence Intervals for Means	3. Practice
4. Proportions and Sample-size Planning	4. Practice

Finish the integrated review. Instructor answers are kept separately.

Cases, readings, and evidence

The main textbook is Anderson et al. (2020), 8th edition. Selected Chapter 7 material develops sampling and sampling distributions; selected Chapter 8 material develops mean and proportion intervals and planning.

- EAI employees: population summaries p.308; sample summaries p.316; finite correction and CLT pp.323–324.
- Lloyd's spending: known-SD inputs p.357.
- Household credit-card balances: rounded sample summaries and t interval pp.367–368.
- Women golfers: survey counts and normal interval p.378; planning pp.380–381.

We use stated book parameters and summaries rather than introducing unverified observation files. Artificial simulations and original planning extensions are labeled. No new workbook-to-CSV match is claimed.

Next step

Module 5: Hypothesis testing and group comparisons follows interval estimation; its existing lesson remains retained material pending expansion.

Return to course home.

Module 4: Sampling and Variability

Lesson 1 | From a population to a sample

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Learning objectives and reading

Distinguish a target population, sampling frame, sample, parameter, and statistic; explain a simple random sample; and separate sampling variability from selection bias. Read main textbook §§7.1–7.4, especially printed pp.308 and 316.

Business case: EAI employees

The textbook's EAI population contains **2,500 employees**, with annual salary mean **$71,800**, population SD **$4,000**, and **1,500 employees** who completed management training. Its sample of **30 employees** has mean salary **$71,814** and **19 training completers**. How much can a sample estimate vary? We use the printed population parameters and sample summaries; the individual personnel records have not been verified for this lesson.

يتكون مجتمع موظفي **EAI** في الكتاب من **2,500 موظف**، بمتوسط راتب سنوي **71,800 دولار** وانحراف معياري للمجتمع **4,000 دولار**، وقد أكمل **1,500 موظف** التدريب الإداري. يبلغ متوسط الراتب في عينة الكتاب المكونة من **30 موظفًا 71,814 دولارًا**، وأكمل **19 منهم** التدريب. إلى أي مدى قد يختلف تقدير العينة؟ نستخدم معلمات المجتمع وملخصات العينة المنشورة؛ ولم نتحقق من سجلات الموظفين الفردية لهذا الدرس.

Define what is sampled

The **target population** is the group about which a conclusion is intended. The **sampling frame** is the list or mechanism from which units can actually be selected. Missing workers in a frame can create coverage bias even if selection from that frame is random. An observational unit here is one employee; salary is measured in dollars per year and training status is categorical.

A **parameter** describes a population, such as μ or p. A **statistic** is computed from a sample, such as $\bar{x}$ or $\hat{p}$. A statistic used to estimate a parameter is an estimator; its realized numerical result is an estimate.

A **simple random sample without replacement** gives every possible subset of n employees the same probability of selection. Equal selection probabilities for individual employees alone do not establish this stronger property. A selected employee cannot appear twice within a sample.

```
# sample(): randomly selects values; size sets the count and replace=FALSE prevents
 ↪  repeats within a draw.
# seq_len(): returns integer positions 1 through n, safely returning an empty vector when
 ↪  n=0.
# head(): shows the first n entries or rows, six by default.
# Select employee identifiers only: no salaries or training outcomes are fabricated.
# set.seed() reproduces the draw; replace=FALSE prevents duplicate IDs within this sample.
set.seed(307312)
selected_ids <- sample(seq_len(2500), size = 30, replace = FALSE)
head(selected_ids)
```

```
[1] 1061 1210 2290  607 2272   11
```

These are illustrative selected IDs, not a reconstruction of the book's sample. Without the checked personnel records, they cannot be joined to verified salaries.

Point estimates and realized errors

```
# kable(): formats a data frame or matrix as a readable report table; digits sets
 ↪  displayed precision.
# data.frame(): builds a table whose named columns have compatible lengths.
# c(): combines values into a vector, converting them to a common type when needed.
# Calculate estimates from the book's summaries, not from artificial employee
 ↪  observations.
# The proportion denominator is all 30 sampled employees; multiplying by 100 gives
 ↪  percentage points.
mean_estimate <- 71814
proportion_estimate <- 19 / 30
knitr::kable(data.frame(
  Quantity = c("Salary estimate (USD/year)", "Salary error (USD/year)",
               "Training estimate", "Training error (percentage points)"),
  Value = c(mean_estimate, mean_estimate - 71800,
            proportion_estimate, 100 * (proportion_estimate - .60))), digits = 4)
```

Quantity	Value
Salary estimate (USD/year)	71814.0000
Salary error (USD/year)	14.0000
Training estimate	0.6333
Training error (percentage points)	3.3333

The realized salary error is $14 and the training error is about 3.33 percentage points. We can calculate these errors only because the teaching example supplies the population parameters. In an actual estimation problem the unknown parameter prevents us from observing the estimation error directly.

Why estimates change

A second random sample generally includes different employees, producing a different statistic. This is **sampling variability**. It does not mean either sample was selected incorrectly. A **sampling distribution** is the distribution of a statistic across repeated samples under the same design, not the distribution of employees' salaries in one sample.

Probability sampling permits design-based uncertainty calculations. A convenience sample of available staff can miss systematic differences; a larger convenience sample need not remove that bias. Nonresponse and measurement errors are further sources of error. Random selection supports generalization to a covered population; random assignment to treatments supports causal comparisons. They serve different purposes.

Other designs include stratified sampling (sample within each stratum), cluster sampling (select groups, then observe all or some units in them), and systematic sampling (a random start followed by regular spacing). Their uncertainty calculations depend on the design. Do not apply the simple-random-sample SE automatically to clustered observations.

Check your understanding

Explain why a sample estimate that differs from its parameter is not automatically biased. Then explain why randomly selecting 30 people from an incomplete employee list may still misrepresent all employees.

Source and limits

Anderson et al. (2020), 8th ed., printed p.308 (population) and p.316 (Table 7.2 and sample summaries). The book reports the sample SD as approximately $3,348; no raw-row compatibility is claimed here. Selection of IDs above is an original course demonstration, not publisher observations.

Complete the practice, then return to the module overview.

Practice: Sampling and Variability

Module 4 | Lesson 1 | From a population to a sample

Use the lesson for the checked case introductions and source limits. Submit a short Quarto report with calculations, units, assumptions, and interpretation.

1. Identify the population, frame, observational unit, parameters, and statistics in EAI.
2. Reproduce the sample salary error and training error in percentage points.
3. Draw 50 distinct IDs from 1:2500 with a fixed seed. Explain `replace = FALSE` and why a different seed may change the result.
4. Explain why we cannot calculate a verified mean salary from these IDs alone.
5. A manager surveys only staff who volunteer. Does increasing the volunteer count guarantee removal of bias? Distinguish this issue from sampling variability.

Module 4: Sampling Distributions and Standard Errors

Lesson 2 | Individuals versus statistics

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Learning objectives and reading

Calculate and interpret standard errors for a mean and a proportion, explain the central limit theorem, and recognize when finite population correction matters. Read §§7.5–7.6, especially printed pp.322–324.

Business case: repeated EAI samples

Return to the textbook's **2,500 EAI employees**, whose annual salaries have mean **$71,800** and population SD **$4,000**; **60%** completed management training. Compare repeated simple random samples of **30** employees. How variable are sample mean salary and sample training proportion? These are stated population summaries, not verified individual records.

نعود إلى مجتمع **2,500 موظف** لدى **EAI** في الكتاب، حيث متوسط الراتب السنوي **71,800 دولار** والانحراف المعياري للمجتمع **4,000 دولار**، وقد أكمل **60%** التدريب الإداري. نقارن عينات عشوائية بسيطة متكررة، حجم كل منها **30 موظفًا**. ما مقدار تباين متوسط الراتب ونسبة إكمال التدريب بين العينات؟ هذه ملخصات المجتمع المعطاة، وليست سجلات فردية جرى التحقق منها.

Three different distributions

The population distribution describes individual salaries; a sample distribution describes the observed salaries in one sample; the sampling distribution describes one statistic over repeated samples. Their horizontal axes may share units, but their spreads answer different questions.

For independent, identically distributed observations with finite variance:

$$E(\bar{X}) = \mu, \qquad SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

The population SD describes variability of individual values. The **standard error** is the SD of an estimator's sampling distribution. When σ is unknown, $s/\sqrt{n}$ estimates the SE; it does not become a known population quantity.

For independent binary observations with common probability p:

$$E(\hat{p}) = p, \qquad SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}.$$

For simple random sampling **without replacement** from a finite population of size N, multiply these SEs by $\sqrt{(N-n)/(N-1)}$ when population variance uses denominator N. Without-replacement observations are dependent. The textbook neglects this correction when $n/N \leq .05$; this is an approximation, not exact independence.

```
# kable(): formats a data frame or matrix as a readable report table; digits sets
 ↪  displayed precision.
# data.frame(): builds a table whose named columns have compatible lengths.
# c(): combines values into a vector, converting them to a common type when needed.
# Compare the book's large-population approximation with the finite-population SE.
# sqrt() takes a square root; the correction uses N-1 because population variance uses
 ↪  denominator N.
N <- 2500
n <- 30
fpc <- sqrt((N - n)/(N - 1))
knitr::kable(data.frame(Statistic = c("Mean salary (USD/year)", "Training proportion"),
 Approximate_SE = c(4000/sqrt(n), sqrt(.6*.4/n)),
 Corrected_SE = fpc * c(4000/sqrt(n), sqrt(.6*.4/n))), digits = 5)
```

Statistic	Approximate_SE	Corrected_SE
Mean salary (USD/year)	730.29674	726.04696
Training proportion	0.08944	0.08892

The uncorrected salary SE is about $730.30, much smaller than the $4,000 individual SD. Multiplying sample size by four halves the uncorrected SE. Bias is not in this formula; smaller SE does not establish accuracy under a faulty design.

Shape and the central limit theorem

For independent normal observations, the sample mean is exactly normal for any n. For many nonnormal populations with finite variance and appropriate independence, the **central limit theorem** makes the mean approximately normal as n grows. It does not make the raw observations normal. There is no universal sample size of 30 that handles arbitrary skewness, outliers, or dependence.

For a proportion, assess expected success and failure counts before a normal approximation. We use $np \geq 10$ and $n(1-p) \geq 10$ as a cautious teaching rule; this is a guideline, not a proof. Use estimated counts when constructing an interval with unknown p.

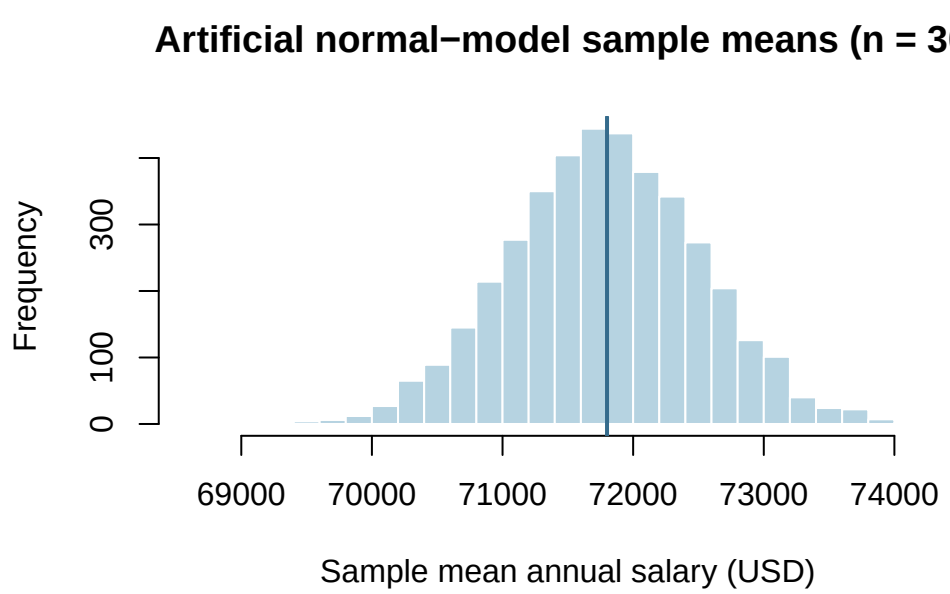
An explicitly artificial illustration

The next simulation uses independent normal salaries with EAI's stated mean and SD. It is an **original model illustration**: no claim is made that the finite EAI salary population is normal or that these are its actual employees. We deliberately omit finite population correction because this simulation draws from an independent model.

```r
# mean(): computes an average; logical TRUE/FALSE values contribute 1/0, giving a
 ↪  proportion.
# c(): combines values into a vector, converting them to a common type when needed.
# sd(): computes sample standard deviation using denominator n-1.
# sqrt(): returns the nonnegative square root of a nonnegative number.
# replicate(B, expression) repeats a calculation B times; each result here is a new sample
 ↪  mean.
# rnorm(n, mean, sd) draws n artificial normal values using SD, not variance; set.seed()
 ↪  fixes the draw sequence.
set.seed(307312)
means <- replicate(4000, mean(rnorm(30, mean = 71800, sd = 4000)))
c(simulated_mean = mean(means), simulated_SE = sd(means), model_SE = 4000/sqrt(30))
```

```
simulated_mean   simulated_SE       model_SE
    71783.2911       730.5734       730.2967
```

```r
# hist() displays the simulated sample means; breaks=30 requests roughly 30 histogram
 ↪  classes.
# abline(v=...) adds a vertical reference at the model's population mean.
hist(means, breaks = 30, col = "#B6D3E1", border = "white",
 main = "Artificial normal-model sample means (n = 30)", xlab = "Sample mean annual salary
  ↪  (USD)")
abline(v = 71800, col = "#356B8C", lwd = 2)
```



Original R simulation from a stated normal model; bars represent sample means, not individual employees.

Interpret a probability

Under that normal model, the chance a sample mean of 30 lies within $500 of $71,800 is:

```r
# sqrt(): returns the nonnegative square root of a nonnegative number.
# pnorm() returns a normal lower-tail probability; subtract the two CDF values for an
 ↪  interval.
# sd receives the SE of a mean, not the $4,000 SD of individual salaries.
se <- 4000 / sqrt(30)
pnorm(72300, mean = 71800, sd = se) - pnorm(71300, mean = 71800, sd = se)
```

```
[1] 0.5064372
```

This is a probability about a future sample mean under the model. It is not a probability that one employee earns within $500 of the mean.

Sources and limits

Anderson et al. (2020), §§7.5–7.6; EAI parameters p.308, finite correction and SE p.323, normality and CLT p.324. Formula use is conditional on the sampling design. Normal simulated salaries are course-generated model values, not reconstructed book data.

Complete the practice, then return to the module overview.

Practice: Sampling Distributions and Standard Errors

Module 4 | Lesson 2 | Individuals versus statistics

Use the lesson for the checked case introductions and source limits. Submit a short Quarto report with calculations, units, assumptions, and interpretation.

1. For EAI, calculate the approximate salary SE at n = 30 and n = 120. What stays unchanged about individual salary SD?
2. Compute finite-population-corrected salary SEs at both sizes. Which sampling fraction exceeds 5%?
3. Calculate the approximate training-proportion SE at n = 30 and check the expected counts for a normal approximation.
4. Repeat the lesson's artificial mean simulation with n = 120. Label its model and compare empirical SD with theoretical SE.
5. Explain why the CLT does not justify a normal salary histogram or fix dependence between coworkers.

Module 4: Confidence Intervals for Means

Lesson 3 | Quantify uncertainty about a mean

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Learning objectives and reading

Construct and interpret a confidence interval for a population mean; distinguish known population SD from estimated SD; and justify a t critical value. Read §§8.1–8.2, especially printed pp.357 and 367–368.

Business case: spending at Lloyd’s

Lloyd’s Department Store samples **100 customers** and reports mean spending **$82 per shopping trip**. The textbook treats the population SD as known at **$20**, supported by historical data, and assumes a normal spending population. What interval estimates population mean spending? We use the stated summaries, not verified individual transactions.

يأخذ متجر **Lloyd’s Department Store** عينة من **100 عميل**، ويبلغ متوسط الإنفاق **82 دولارًا لكل رحلة تسوق**. يعامل الكتاب الانحراف المعياري للمجتمع على أنه معلوم ويساوي **20 دولارًا** استنادًا إلى بيانات تاريخية، ويفترض أن الإنفاق في المجتمع يتبع توزيعًا طبيعيًا. ما الفترة التي تقدّر متوسط الإنفاق في المجتمع؟ نستخدم الملخصات المعطاة، وليس معاملات فردية جرى التحقق منها.

From an estimator to an interval

A point estimate alone omits uncertainty. For a two-sided confidence level $1-\alpha$, a known-SD normal interval is

$$\bar{x} \pm z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}.$$

Here $z_{1-\alpha/2}$ is the standard normal **lower-tail quantile**. At 95% confidence, $\alpha = .05$ and the quantile is about 1.96. The margin of error is the critical value times SE; interval width is twice the margin.

```
# sqrt(): returns the nonnegative square root of a nonnegative number.
# kable(): formats a data frame or matrix as a readable report table; digits sets
 ↪  displayed precision.
# data.frame(): builds a table whose named columns have compatible lengths.
# qnorm(.975) leaves .025 in each normal tail for a two-sided 95% interval.
# c(-1,1) subtracts and adds the margin of error to give lower and upper endpoints in
 ↪  dollars.
se <- 20 / sqrt(100)
margin <- qnorm(.975) * se
ci <- 82 + c(-1, 1) * margin
knitr::kable(data.frame(Estimate = 82, SE = se, Margin = margin, Lower = ci[1], Upper =
 ↪  ci[2]), digits = 3)
```

Estimate	SE	Margin	Lower	Upper
82	2	3.92	78.08	85.92

The interval is approximately **$78.08 to $85.92** for population mean spending per trip. It is not an interval containing 95% of customer spending, nor a prediction interval for one customer.

What 95% confidence means

In repeated samples under the assumptions, about 95% of intervals constructed by this procedure contain the fixed population mean. After observing one interval, that interval either contains the mean or does not. The frequentist confidence level describes the procedure's long-run coverage, not a posterior probability assigned to a fixed parameter.

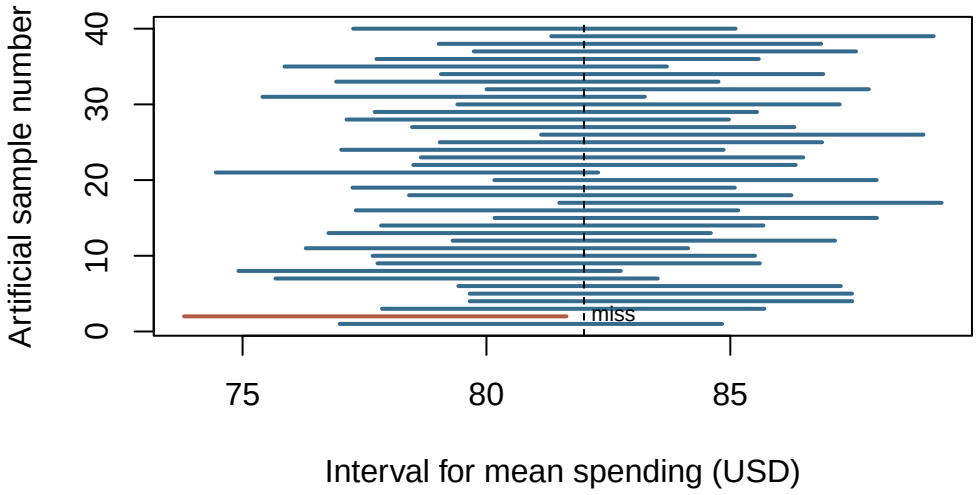
The following illustration fixes an artificial population mean at $82 solely to demonstrate coverage. The actual Lloyd's population mean is unknown.

```
# set.seed(): fixes the pseudorandom sequence so rerunning the same code reproduces its
 ↪  artificial draws.
# rnorm(): generates artificial normal draws; n is the draw count and sd is standard
 ↪  deviation.
# qnorm(): returns a normal lower-tail quantile; sd is standard deviation, not variance.
# Simulate sample means directly: under the normal model their SD is 20/sqrt(100).
# A logical containment test is TRUE for covered intervals; mean() turns TRUE/FALSE into a
 ↪  coverage fraction.
set.seed(307312)
centers <- rnorm(1000, mean = 82, sd = 2)
lower <- centers - qnorm(.975) * 2
upper <- centers + qnorm(.975) * 2
mean(lower <= 82 & upper >= 82)
```

```
[1] 0.952
```

```
# range(): returns the minimum and maximum as a two-element vector.
# c(): combines values into a vector, converting them to a common type when needed.
# ifelse(): chooses a value element by element according to a logical condition.
# text(): adds labels to a base plot; pos sets the side and cex scales label size.
# plot(type="n") sets up empty axes; segments() draws the first 40 simulated intervals.
# Red intervals miss the known artificial mean, marked by abline(v=82); labels also
 ↪  identify misses.
keep <- 1:40
miss <- lower[keep] > 82 | upper[keep] < 82
plot(range(c(lower[keep],upper[keep])), c(1,40), type="n",
```

```
 xlab="Interval for mean spending (USD)", ylab="Artificial sample number")
segments(lower[keep], keep, upper[keep], keep, col=ifelse(miss,"#B35C44","#356B8C"),
 ↪  lwd=2)
text(upper[keep][miss], keep[miss], labels=" miss", pos=4, cex=.7)
abline(v=82, lty=2)
```



Original artificial coverage illustration; intervals vary across samples and the model mean stays fixed.

Business case: household credit-card balances

The textbook's sample of **70 U.S. households** gives mean credit-card balance **$9,312** and sample SD **$4,007**. Population SD is unknown. Estimate the population mean balance with 95% confidence. We use the book's rounded sample summaries; the individual household records and sampling implementation are not independently checked here.

تعطي عينة الكتاب المكونة من **70 أسرة أمريكية** متوسط رصيد بطاقة ائتمان **9,312 دولارًا** وانحرافًا معياريًا للعينة **4,007 دولارات**. الانحراف المعياري للمجتمع غير معلوم. نريد تقدير متوسط الرصيد في المجتمع بثقة 95%. نستخدم ملخصات العينة المدوّرة في الكتاب؛ ولم نتحقق هنا بصورة مستقلة من سجلات الأسر الفردية أو تنفيذ أسلوب المعاينة.

Unknown SD: use a t interval

Estimating σ introduces uncertainty. For independent normal observations,

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}, \qquad \bar{x} \pm t_{1-\alpha/2,n-1}\frac{s}{\sqrt{n}}.$$

The t distribution is symmetric with heavier tails than the standard normal. Its degrees of freedom are $n - 1$ because deviations from the fitted sample mean sum to zero. At larger degrees of freedom its critical values approach normal critical values. **Unknown population SD calls for t even when n exceeds 30.**

For nonnormal data this interval may be approximately valid with a sufficiently informative sample; severe skewness, extreme outliers, dependence, or a poor sampling design require further

assessment. A summary alone cannot reveal those features. Exact normal-theory coverage is conditional on independent normal observations.

```
# sqrt(): returns the nonnegative square root of a nonnegative number.
# c(): combines values into a vector, converting them to a common type when needed.
# kable(): formats a data frame or matrix as a readable report table; digits sets
 ↪  displayed precision.
# data.frame(): builds a table whose named columns have compatible lengths.
# qt(.975,df=69) gives a two-sided 95% t critical value with n-1 degrees of freedom.
# Use the sample SD in the estimated SE; rounded book summaries produce approximate
 ↪  endpoints.
n <- 70
xbar <- 9312
s <- 4007
critical <- qt(.975, df = n - 1)
margin <- critical * s/sqrt(n)
ci <- xbar + c(-1,1) * margin
knitr::kable(data.frame(df=n-1, Critical=critical, SE=s/sqrt(n), Margin=margin,
 Lower=ci[1], Upper=ci[2]), digits=3)
```

df	Critical	SE	Margin	Lower	Upper
69	1.995	478.928	955.435	8356.565	10267.43

The interval is about **$8,357 to $10,267**, matching the book to rounded dollars. This inference is conditional on the sample supporting generalization to the stated household population. If raw data were available, `t.test(x, conf.level = .95)$conf.int` would compute a one-sample t interval; the vector x must contain observations, not the three summary values.

How precision changes

Higher confidence requires a larger critical value and a wider interval for the same sample. Larger n generally narrows the interval under the same design and variability. Neither change repairs bias. If sampling without replacement takes more than 5% of a finite population, revisit design-specific SEs rather than using these large-population formulas automatically.

Sources and limits

Anderson et al. (2020), §8.1, p.357 (Lloyd's inputs); §8.2, pp.367–368 (household inputs and interval). Historical SD is treated as known only under the book's modeling assumption. Household summaries are rounded; no workbook match or independently checked distribution shape is asserted.

Complete the practice, then return to the module overview.

Practice: Confidence Intervals for Means

Module 4 | Lesson 3 | Quantify uncertainty about a mean

Use the lesson for the checked case introductions and source limits. Submit a short Quarto report with calculations, units, assumptions, and interpretation.

1. Reproduce the 95% Lloyd's interval and calculate a 99% interval using the same summaries. Compare widths.
2. Explain why neither interval describes the range of individual customer spending.
3. Reproduce the 95% household-balance t interval. Calculate a 90% interval; identify df, estimated SE, and margin.
4. Explain why using z merely because n = 70 is inappropriate when population SD is unknown.
5. Explain 95% confidence without assigning 95% probability to the fixed population mean. Name two assumptions that summaries alone cannot verify.

Module 4: Proportions and Sample-size Planning

Lesson 4 | Estimate a share and plan precision

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Learning objectives and reading

Construct a large-sample proportion interval, check its conditions, and plan a sample size for a proportion or mean. Read §§8.3–8.4, especially printed pp.378–381. The proportion interval below reproduces the book's normal approximation; it is not a universal method for small counts.

Business case: women golfers and tee times

The textbook reports a national survey of **900 women golfers**, of whom **396** were satisfied with the availability of tee times. What interval estimates the population proportion satisfied? How many completed responses would a new survey need for a margin of error of **0.025** at **95% confidence**? We use the published counts and planning question, not verified individual responses or an independently checked survey design.

يعرض الكتاب مسحًا وطنيًا شمل **900 لاعبة غولف**، كانت **396 منهن** راضيات عن توافر مواعيد بدء اللعب. ما الفترة التي تقدّر نسبة الراضيات في المجتمع؟ وكم استجابة مكتملة يحتاجها مسح جديد لهامش خطأ **0.025** عند مستوى ثقة **95**؟ نستخدم الأعداد وسؤال التخطيط المنشورين، وليس استجابات فردية جرى التحقق منها أو تصميم مسح تحقّقنا منه بصورة مستقلة.

From a binary response to a proportion

Define success before counting: here it means **satisfied**, not every favorable response on an unspecified question. For x successes among n responses,

$$\hat{p} = \frac{x}{n}, \qquad \widehat{SE} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

$$\hat{p} \pm z_{1-\alpha/2}\widehat{SE}.$$

This is the normal or **Wald interval**. It assumes a design supporting independent binary responses or an appropriate small sampling fraction, and a meaningful target population. Check $n\hat{p} \geq 10$ and $n(1-\hat{p}) \geq 10$ in this course. A large count does not establish representative selection or eliminate nonresponse bias.

```
# c(): combines values into a vector, converting them to a common type when needed.
# kable(): formats a data frame or matrix as a readable report table; digits sets
 ↪  displayed precision.
# data.frame(): builds a table whose named columns have compatible lengths.
# The success denominator includes all 900 surveyed women golfers, not only the satisfied
 ↪  respondents.
# qnorm(.975) is the two-sided 95% normal critical value; sqrt() computes the estimated
 ↪  SE.
x <- 396
n <- 900
phat <- x/n
se <- sqrt(phat*(1-phat)/n)
margin <- qnorm(.975)*se
ci <- phat+c(-1,1)*margin
knitr::kable(data.frame(Estimate=phat, Successes=x, Failures=n-x,
 SE=se, Margin=margin, Lower=ci[1], Upper=ci[2]),digits=5)
```

Estimate	Successes	Failures	SE	Margin	Lower	Upper
0.44	396	504	0.01655	0.03243	0.40757	0.47243

Both counts (396 and 504) are comfortably above 10. The interval is approximately **0.4076 to 0.4724**, or **40.76% to 47.24%**. The margin is about **3.24 percentage points**, not 3.24% of the estimated proportion.

If there are very few successes or failures, the Wald approximation may fail badly, including zero width or endpoints outside [0,1]. Do not silently clip endpoints and claim that coverage is repaired. Score or exact binomial methods are possible extensions. R's default `prop.test()` uses a score-based approach with a continuity correction, so its interval need not equal this textbook Wald calculation.

Plan a proportion sample size

Before collecting new data, choose confidence, desired margin E, and a planning value p^*:

$$n = \left\lceil \frac{z_{1-\alpha/2}^2 p^*(1-p^*)}{E^2} \right\rceil.$$

The ceiling rounds **up** because the minimum required size must be met. A prior survey can suggest p^* but cannot guarantee the new population share. If no defensible value is available, $.50$ maximizes $p^*(1-p^*)$ and is conservative within this simple-sampling normal approximation.

```
# qnorm(): returns a normal lower-tail quantile; sd is standard deviation, not variance.
# c(): combines values into a vector, converting them to a common type when needed.
# kable(): formats a data frame or matrix as a readable report table; digits sets
 ↪  displayed precision.
# data.frame(): builds a table whose named columns have compatible lengths.
# ceiling() rounds upward to a whole number of completed responses; E=.025 means 2.5
 ↪  percentage points.
```

```r
# Compare the prior-survey planning share with .50; exact R quantiles may differ slightly
 ↪  from rounded tables.
z <- qnorm(.975)
E <- .025
planning_p <- c(.44,.50)
required <- ceiling(z^2*planning_p*(1-planning_p)/E^2)
knitr::kable(data.frame(Planning_share=planning_p, Completed_responses=required))
```

Planning_share	Completed_responses
0.44	1515
0.50	1537

The required completed samples are **1,515** using .44 and **1,537** using .50, matching the book's recommendations. This controls the planned normal-approximation margin under the design assumptions. It does not guarantee response or remove coverage bias. Invitations must exceed completed responses if some invitees do not respond; nonresponse adjustment requires a justified response-rate assumption.

Plan precision for a mean

With a defensible planning SD σ^*, the normal-based planning formula is

$$n = \left\lceil \left(\frac{z_{1-\alpha/2}\sigma^*}{E} \right)^2 \right\rceil.$$

Use E and σ^* in the **same measurement units**. A pilot or relevant historical source may provide the planning SD. This is a planning approximation when SD is estimated; a final unknown-SD mean interval still uses t. Complex designs and substantial sampling fractions need their corresponding design calculations.

For an **original planning extension** of the checked Lloyd's example, take its $20 SD and request a $2 margin at 95% confidence:

```r
# qnorm(): returns a normal lower-tail quantile; sd is standard deviation, not variance.
# Use the checked Lloyd's SD in an explicitly new planning question, not a new observed
 ↪  sample.
# Squaring gives required n; ceiling() preserves the requested maximum planned margin.
ceiling((qnorm(.975)*20/2)^2)
```

```
[1] 385
```

This gives 385 completed observations under the stated assumptions. Halving E roughly quadruples the required sample size. More precision has a cost; select a practically useful margin before collecting data.

Choose the method after learning the conditions

Target and inputs	Interval or planning tool	Key check
Mean, known population SD	Normal z interval	SD genuinely treated as known; valid design
Mean, unknown population SD	t interval with n−1 df	Outliers, shape, independence, sampling design
Binary proportion, adequate counts	Textbook normal/Wald interval	Success/failure counts and representative design
Planned mean precision	z-based planning SD formula	Units and credibility of SD
Planned proportion precision	Planning share formula	E in proportion units; round up

Sources and limits

Anderson et al. (2020), §8.4, p.378 (396/900 and interval), pp.380–381 (E=.025 and sample sizes). Mean planning formula: §8.3; the Lloyd's $2 margin is an explicitly original extension using the SD on p.357. Published survey counts do not verify response order, sampling implementation, or present-day satisfaction.

Complete the practice, then return to the module overview.

Practice: Proportions and Sample-size Planning

Module 4 | Lesson 4 | Estimate a share and plan precision

Use the lesson for the checked case introductions and source limits. Submit a short Quarto report with calculations, units, assumptions, and interpretation.

1. Reproduce the 95% tee-time Wald interval and interpret its margin in percentage points.
2. Calculate a 90% interval from the same counts. Compare widths without treating them as different datasets.
3. Reproduce both 95% planning sample sizes for E=.025. Then halve E and compare the unrounded required sizes.
4. Explain why ceiling(), not round(), is required and why p*=.50 is conservative in this formula.
5. For Lloyd's original planning extension, find n for a $1 margin at 95% confidence. State the role and units of the planning SD.
6. Explain why default prop.test() may give different limits from the lesson and why large n alone cannot fix nonresponse bias.

Module 4: Sampling and Estimation Review

Select, calculate, interpret, qualify

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Bridge to hypothesis testing 1

Integrated tasks

Use the checked cases from Lessons 1–4; no new business observations are introduced. Submit a concise report with a method choice, calculation, interpretation, and limitation for each task.

1. **Selection and bias:** Explain why a random sample from an incomplete EAI employee frame may not represent all employees. Distinguish parameter, estimate, realized error, and SE.
2. **Sampling distribution:** For EAI salary samples of n=100, compute the approximate and corrected SE for N=2500. Contrast the SE with the $4,000 population SD.
3. **Mean interval:** Use the household summaries n=70, mean=$9,312, s=$4,007 to construct a 95% interval. Explain the critical distribution and df.
4. **Proportion interval:** Use 396/900 for tee-time satisfaction. Compute a 95% Wald interval and state its conditions. Explain why it describes a population proportion rather than a range of binary observations.
5. **Planning:** Find the completed sample needed for E=.025 and p*=.50 at 95% confidence. Identify what the result does and does not guarantee.
6. **Communication:** Correct these claims: “95% of salaries lie in the mean confidence interval”; “n>30 means use z”; “a huge volunteer sample removes bias”; “the CLT makes the population normal.”
7. **Evidence limits:** List which inputs are published summaries, which computations rely on assumptions, and what raw data or design evidence would be required to assess those assumptions.

Bridge to hypothesis testing

An interval shows plausible parameter values under a procedure and its assumptions. The next module evaluates a specified null claim. Retain the same target parameter, sampling conditions, and distinction between practical importance and statistical uncertainty.

Return to the module overview.